NARAYANA
IIT ACADEMY
INDIA

JEE ADVANCED
GRAND TEST SERIES

NARAYANA
IIT ACADEMY
INDIA

Sec: OSR.IIT_*CO-SC	**Date: 18-04-24**
Time: 3HRS	**Max. Marks: 180**

Name of the Student: ____________________ **H.T. NO:** ☐☐☐☐☐☐☐

18-04-24_OSR.STAR CO-SUPER CHAINA_JEE-ADV_GTA-6(P2)_SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

MATHEMATICS: TOTAL SYLLABUS

TIME: 3HRS **IMPORTANT INSTRUCTIONS** Max Marks: 180

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 6)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II (Q.N : 7 – 12)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 13 – 16)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	+3	–1	4	12
Sec – IV (Q.N : 17 – 19)	Questions with Non-Negative Integer type	+4	0	3	12
Total				**19**	**60**

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 20 – 25)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II (Q.N : 26 – 31)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 32 – 35)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	+3	–1	4	12
Sec – IV (Q.N : 36 – 38)	Questions with Non-Negative Integer type	+4	0	3	12
Total				**19**	**60**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 39 – 44)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II (Q.N : 45 – 50)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 51 – 54)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	+3	–1	4	12
Sec – IV (Q.N : 55 – 57)	Questions with Non-Negative Integer type	+4	0	3	12
Total				**19**	**60**

MATHEMATICS MAX.MARKS: 60

SECTION - 1 (Maximum Marks : 24)

This section contains **SIX (06)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

1. ΔABC, whose vertices are $A(3,-2), B(3,1) \& C(1,5)$.P is a point on altitude AD of ΔABC such that ordinate of P is 5. Now a circle is drawn taking P as centre, DE as one of the tangent, where E is the point on AB such that AB is perpendicular to CE. Identify the true statement.

A) If tangents are drawn from point $D \& E$ to given circle, then they intersect at a point on AC.

B) PDBE is a cyclic quadrilateral.

C) Centre of given circle is $(17,5)$

D) $\angle PBD = \angle DEP$

2. Consider a circle $x^2 + y^2 = 25$ with centre P_1. Pair to tangents are drawn from $P_2\left(-\frac{25}{3}, 0\right)$ to given circle which touch the circle at $T_1 \& T_2$. If 2 circles each of radius 3 are drawn with centre $C_1 \& C_2$ which touch the given circle externally at $T_1 \& T_2$ respectively, then

A) Area of $\Delta P_1C_1C_2$ is $\frac{786}{25}$

B) Area of $\Delta P_2C_1C_2$ is $\frac{1969}{75}$

C) Area of $\Delta P_1C_1C_2$ is $\frac{768}{25}$

D) Area of $\Delta P_2C_1C_2$ is $\frac{1696}{75}$

3. Let the number of ways in which 3 positive integers a,b,c can be selected such that $abc = 30030$ is k, then

A) If a,b,c be not necessarily distinct, then value of k is 726 .

B) If $a < b < c$, then value of k is 121 .

C) Value of k is 729 if a,b,c be not necessarily distinct.

D) If $a < b < c$, then value of k is 115 .

4. Consider a parabola whose directrix passes through $P(0,9)$. If $T(3,2)$ is the foot of perpendicular drawn from focus $S(2,-1)$ on a tangent of parabola, then -

A) Length of latus rectum of parabola is $8\sqrt{2}$

B) Equation of tangent at vertex is $x+y-5=0$

C) Equation of axis of parabola is $x-y=3$

D) Directrix is at a distance $2\sqrt{2}$ from focus.

5. If $f(n,x)=\sum_{k=0}^{n} {}^{n}C_{k}\sin(kx)\cos((n-k)x)$, then

A) $f\left(10,\dfrac{\pi}{20}\right)=512$

B) $f\left(5,\dfrac{\pi}{20}\right)$ satisfies the equation $x^2=256$

C) $f\left(11,\dfrac{\pi}{11}\right)=0$

D) $f\left(7,\dfrac{\pi}{14}\right)=256$

6. If $\left(\sum_{r=1}^{2016}2^{r-1}\tan 2^{r-1}\theta\right)=\cot\dfrac{\pi}{2^{\lambda}}-2^{\mu}$ (where $\theta=\dfrac{\pi}{2^{2018}}$), then identify correct option(s).

A) $\lambda-\mu=2$

B) $\lambda+\mu=(\lambda-\mu)(\mu+1)$

C) $\lambda^{\mu}<\mu^{\lambda}$

D) $\lambda^{\mu}>\mu^{\lambda}$

SECTION - 2 (Maximum Marks : 12)

This section contains **TWO** (**3**) Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +2 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 7 and 8:

Five balls are to be placed in three boxes. Boxes should hold all the five balls so that no box remains empty.

7. Find number of ways if balls and boxes are identical is

8. Find number of ways if balls as well as boxes are identical but boxes are kept in a row is

Question Stem for Question Nos. 9 and 10:

Let A be the set of all eight-digit numbers that can be formed using all the digits 1,1,2,2,3,3,4,5. A number is selected at random from A

9. If the probability such that no two identical digits appear together is λ, then 168λ is:

10. If the probability that exactly two pairs of identical digits appear together is k, then 63k is

Question Stem for Question Nos. 11 and 12:

Given that $a > 0, |ax^2 + bx + c| \le 1$, if $-1 \le x \le 1, a, b, c \in R$ and $ax + b$ has its maximum value 2, when $-1 \le x \le 1$. Then :

11. 'a' will be

12. 'b' will be

SECTION - 3 (Maximum Marks : 12)

This section contains **TWO** (**2**) Paragraphs. Based on each paragraph, there are **2** questions.
Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer. For each question, choose the option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.
Negative Marks: −1 In all other cases.

Question Stem for Question Nos. 13 and 14:

In a bag, there are 10 cards labeled from 1 to 10, 3 cards are drawn, denoted by $a, b, c\,(a > b > c)$ at a time from the bag

13. If $S = \left\{ (a,b,c) / \int_0^a \left(x^2 - 2bx + 3c\right)dx = 0 \right\}$ then $n(S) =$

A) 4 B) 3 C) 2 D) 5

14. Probability that $\int_0^a \left(x^2 - 2bx + 3c\right)dx = 0$ is

A) $\frac{1}{30}$ B) $\frac{1}{40}$ C) $\frac{1}{20}$ D) $\frac{1}{120}$

Question Stem for Question Nos. 15 and 16:

Suppose you do not know the function $f(x)$, however some information about $f(x)$ is listed below. Read the following carefully before attempting the questions.

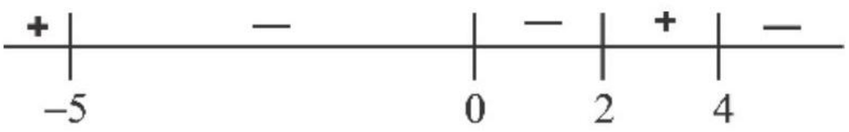
(i) $f(x)$ is continuous and defined for all real numbers

(ii) $f'(-5) = 0; f'(2)$ is not defined and $f'(4) = 0$

(iii) $(-5, 12)$ is a point which lies on the graph of $f(x)$

(iv) $f''(2)$ is undefined, but $f''(x)$ is negative everywhere else.

(v) the signs of f' are given below :



15. On the possible graph of $y=f(x)$ we have

A) $x=-5$ is a point of relative minima

B) $x=2$ is a point of relative maxima

C) $x=4$ is a point of relative minima

D) graph of $y=f(x)$ must have a geometrical sharp corner or cusp

16. From the possible graph of $y=f(x)$, we can say that

A) there is exactly one point of inflection on the curve

B) $f(x)$ increases on $-5<x<2$ and $x>4$ and decreases on $-\infty<x<5$ and $2<x<4$

C) the curve is always concave down

D) curve always concave up

SECTION - 4 (Maximum Marks : 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

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17. Let A be a square matrix such that $A(\text{adj}A)=\begin{bmatrix}4&0&0\\0&4&0\\0&0&4\end{bmatrix}$, then $\dfrac{|\text{adj}(\text{adj}A)|}{4|\text{adj}A|}$ is

18. If z_1,z_2,z_3 are three points lying on the circle $|z|=2$, then minimum value of $|z_1+z_2|^2+|z_2+z_3|^2+|z_3+z_1|^2-4$ is

19. Given that graph of function $f(x)=(ax+\ell nx+1)(x+\ell nx+1), a\in R$ has at least '3' intersection points with the graph of function $g(x)=x^2$. If the complete interval of 'a' satisfying above conditions is (K_1,K_2) then $2(K_1+K_2)$ is

PHYSICS MAX.MARKS: 60

SECTION - 1 (Maximum Marks : 24)

This section contains **SIX (06)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
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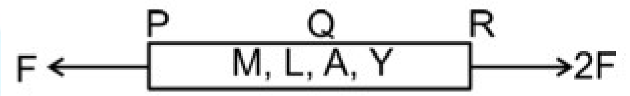
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

20. A uniform rod PR of mass M and length L, area of cross-section A is placed on a smooth horizontal surface. The forces acting on the rod are shown in the figure. Here Y is Young's modulus of elasticity of the rod. Choose the correct options.

(Given that $PQ = QR = \frac{L}{2}$)



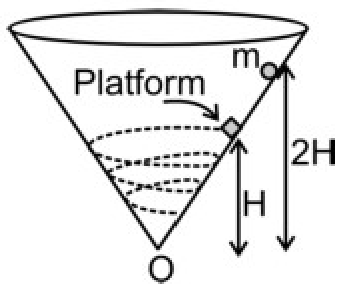
A) Ratio of elongation in section PQ of rod and section QR of rod is $5:7$.

B) Elastic potential energy stored in the PQ section of rod is $\frac{19\ F^2\ L}{48AY}$.

C) Ratio of elastic potential energy stored in section PQ and section QR of the rod is $29:37$.

D) The total elastic potential energy stored in the rod is $\frac{7\ F^2\ L}{6AY}$.

21. A fixed hollow frictionless cone is positioned with its tip pointing down. A small particle of mass m is released from rest on the inside surface from height $2H$ measured from vertex O of the cone. After it has slid half-way down to the tip, it bounces elastically off a platform. The platform is positioned at a 45° angle along the surface of the cone, so that the particle ends up being deflected horizontally along the surface of the cone (into the plane of the diagram shown). The particle then swirls up and around the cone before coming down moving in circular paths and always remaining in contact with the cone's inner surface.



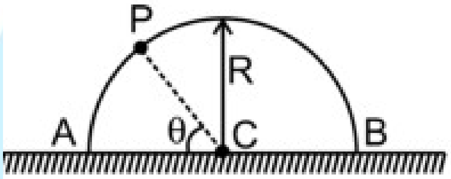
A) The particles maximum swirling height from point ' O ' after its collision with the platform is $\left(\frac{\sqrt{5}+1}{2}\right)H$.

B) The particles maximum swirling height from point ' O ' after its collision with the platform is $\frac{H}{\sqrt{5}+1}$.

C) The speed of the particle at the maximum height from point O after its collision with the platform is $\frac{\sqrt{8gH}}{\sqrt{5}+1}$.

D) The speed of the particle at the maximum height from point O after its collision with the platform is $\left(\sqrt{5}+1\right)\sqrt{8gH}$.

22. A uniform hemispherical bowl of mass ' M ' is at rest on frictionless horizontal ground surface. There is a small insect of mass ' m ' on bowl at point ' A ' of the bowl at rest. Now the insect moves on the hemisphere with constant speed ' v ' relative to bowl in vertical plane. Assume that insect does not slip on the bowl.



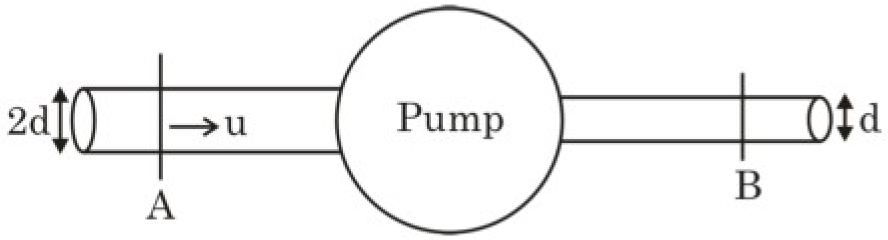
A) When the insect reaches point ' P ' of the hemisphere the displacement of hemisphere w.r.t. ground is $\frac{mR}{(M+m)}(1+\cos\theta)$ horizontally towards left.

B) When the insect is at point ' P ' of the hemispherical bowl the acceleration of bowl is $\left(\frac{m}{M+m}\right)\frac{V^2\cos\theta}{R}$ horizontally towards left.

C) When the insect is at point ' P ' of the hemispherical bowl the acceleration of bowl is $\left(\frac{m}{M+m}\right)\frac{V^2\cos\theta}{2R}$ horizontally towards left.

D) When insect is at point ' P ' on the bowl the displacement of centre of mass of the system (bowl + insect) is $\frac{mR\sin\theta}{M+m}$ vertically upwards.

23. Water is supplied by a pump at a steady rate. Inlet and outlet pipes of common access attached to the pump are horizontal and have diameters of cross-sections 2 d and d respectively as shown in the figure. Assume that flow through the entries system is steady and that the viscous effects are neglected. Velocity of water at the inlet is $u = 1\text{ m/s}$ and pump supplies mechanical energy to water in the direction of flow of water which is equal to 2000 J/m^3.P_A and P_B are pressure at section A and B respectively. Take ρ (density of water) $= 1000\text{ kg/m}^3$. Mark the correct options.



A) Speed of water at the outlet is 4 m/s.

B) Speed of water at the outlet is 1 m/s.

C) $P_A - P_B = 1500\text{ N/m}^2$

D) $P_A - P_B = 5500\text{ N/m}^2$

24. Charges are distributed in a spherical shell with an inner radius R_1 and an outer radius R_2. The charge density in the shell is $\rho = a + br$, where r is the distance from the center to the observation point. There is no charge distribution elsewhere in the space :-

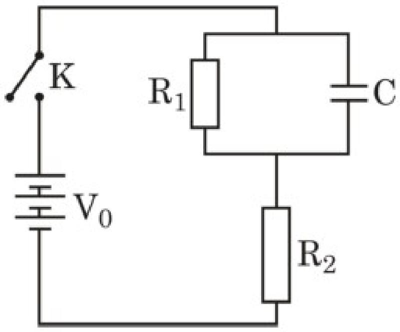
A) The electric field for $R_1 < r < R_2$ is given by $E_2 = \frac{1}{\varepsilon_0 r^2}\left[\frac{a}{3}\left(r^3 - R_1^3\right) + \frac{b}{4}\left(r^4 - R_1^4\right)\right]$.

B) The electric potential for $R_1 < r < R_2$ is zero.

C) The electric potential for $r < R_1$ is $V = \frac{1}{\varepsilon_0}\left[\frac{a}{2}\left(R_2^2 - R_1^2\right) + \frac{b}{4}\left(R_2^3 - R_1^3\right)\right]$.

D) The total charge in the spherical shell is $4\pi\left[\frac{a}{3}\left(R_2^3 - R_1^3\right) + \frac{b}{4}\left(R_2^4 - R_1^4\right)\right]$.

25. A circuit consists of two resistors R_1, R_2 and capacitor C as shown in figure. The capacitor consists of two circular plates with a radius b and separation $d(b >> d)$. The key K is kept closed for a very long time. At $t = 0$, the key K is opened :- $\left(C = \frac{\epsilon_0 \pi b^2}{d}\right)$



A) The magnitude of displacement current just after the key is opened is $\dfrac{V_0}{R_1+R_2}$.

B) The magnitude of magnetic field for $r<b$ is $B=\dfrac{\mu_0 r}{2\pi b^2}\cdot\dfrac{V_0}{R_1+R_2}e^{-\frac{1}{R_1C}t}$.

C) The magnitude of Poynting vector for $r<b$ is $S=\dfrac{R_1V_0^2 r}{2\pi d b^2\left(R_1+R_2\right)^2}e^{-\frac{2}{R_1C}t}$.

D) The magnitude of Poynting vector for $r<b$ is $S=\dfrac{R_2V_0^2 r}{2\pi d^2\left(R_1+R_2\right)}e^{-\frac{2}{R_1C}t}$.

SECTION - 2 (Maximum Marks : 12)

This section contains **TWO** (**3**) Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
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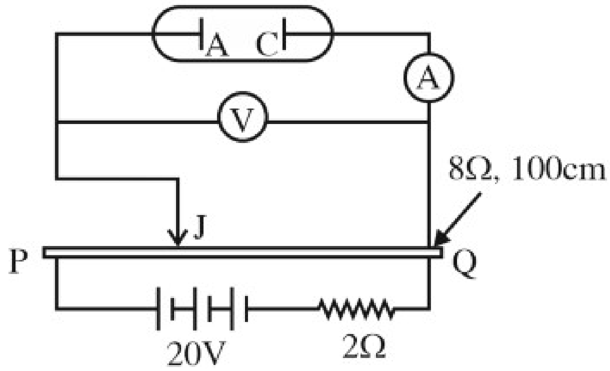
Full Marks: +2 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 26 and 27:

An experimental setup of verification of photoelectric effect is shown in Figure. The voltage across the electrodes is measured with the help of an ideal voltmeter, and which can be varied by moving jockey ' J ' on the potentiometer wire. The battery used in potentiometer circuit is of 20 V and its internal resistance is 2Ω. The resistance of 100 cm long potentiometer wire is 8Ω.



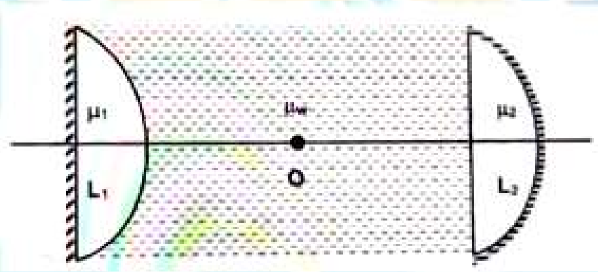
The photocurrent is measured with the help of an ideal ammeter. Two plates of potassium oxide of area $50\ cm^2$ at separation $0.5\ mm$ are used in the vacuum tube. Photocurrent in the circuit is very small, so we can treat the potentiometer circuit as an independent circuit. (Take : $\epsilon_0 = 8.85\times10^{-12} C^2 / N - m^2$)

26. The number of electrons that appear on the surface of the cathode plate are given as $K\times10^9$, when the jockey is connected at the end ' P ' of the potentiometer wire. Assume that no radiation is falling on the plates. Find the K.

27. When a light falls on the anode plate, the ammeter reading remains zero till jockey is moved from the end P to the middle point of the wire PQ. There after, the deflection is recorded in the ammeter. The maximum kinetic energy (in eV) of the emitted electron is :-

Question Stem for Question Nos. 28 and 29:

A cylindrical tube filled with water $(\mu = 4/3)$ is closed at its both ends by two thin silvered plane convex lenses as shown in the figure. Refractive index of lenses L_1 and L_2 are 2.0 and 1.5 while their radii of curvature are 5cm and 9cm respectively. A point object is placed somewhere at a point O on the axis of cylindrical tube. It is found that the object and images coincide with each other.



28. The position of object w.r.t. lens L_1 is____cm

29. The position of object w.r.t. lens L_2 is_____cm

Question Stem for Question Nos. 30 and 31:

Two identical and coaxial superconducting insulated circular coils 1 and 2 with radius R and self inductance L each are located far apart from each other. Each coil carries a current I in the same sense. Now both the coils are moved slowly so that both coils are in contact with each other.

30. The current through each coil in final configuration is α. The value of α is :-

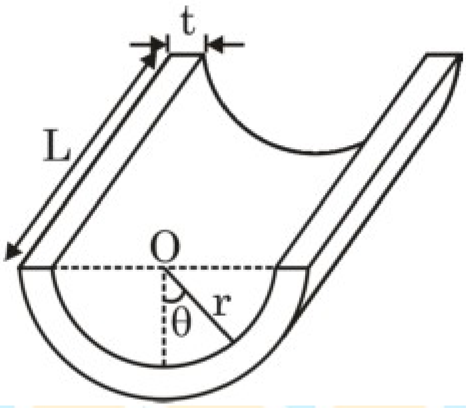
31. The work done by external force in changing the configuration of the system is βLI^2. The value of $|\beta|$ is :-

SECTION - 3 (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.
Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer. For each question, choose the option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.
Negative Marks: −1 In all other cases.

Question Stem for Question Nos. 32 and 33:

A conductor of length L has the shape of a half cylinder of radius r. Thickness of the sheet used to make the conductor is t. The conductivity of the material varies with the angle θ as $\sigma = \sigma_0 \cos\theta$. A battery of emf V and negligible internal resistance is connected across the two semi-circular ends.



32. Which of the following is true regarding the current density (j)?

A) Current density is same at all point on the semi-circular cross-section.

B) Current density varies on the semi-circular cross-section, decreasing with θ.

C) Current density varies on the semi-circular cross-section, increasing with θ

D) Current density is same at every point in the conductor.

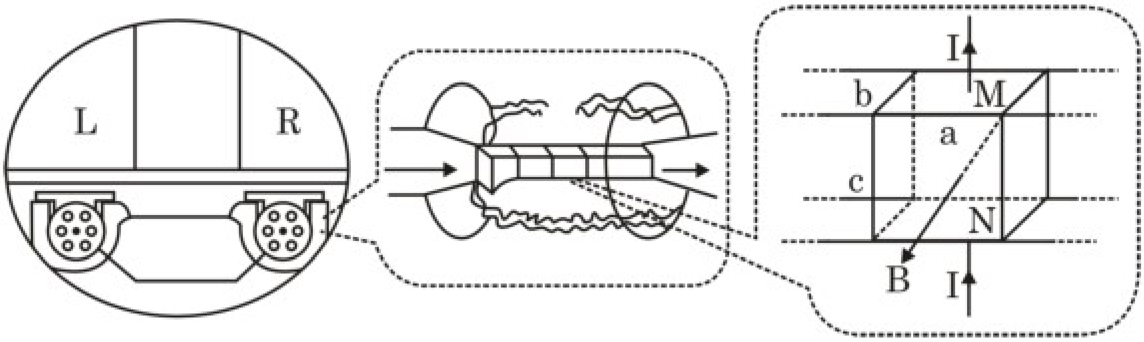
33. Current flowing through the battery is :-

A) $\frac{\sigma_0 \text{ Vrt}}{2\text{ L}}$ B) $\frac{\sigma_0 \text{ Vrt}}{\text{L}}$ C) $\frac{2\sigma_0 \text{VLt}}{\text{r}}$ D) $\frac{2\sigma_0 \text{ Vrt}}{\text{L}}$

Question Stem for Question Nos. 34 and 35:

In order to reduce the noise of the submarine and increase its forward speed, the propeller can be replaced by an electromagnetic propeller. There are two sets of left and right thrusters below the submarine. Each set is composed of 6 identical linear channel thrusters made of insulating materials. The principle diagram is as shown below. The linear channel is filled with seawater with a resistivity of $\rho = 0.2 \text{ohm} - \text{m}$. In the space of $\text{a} \times \text{b} \times \text{c} = 0.3\text{ m} \times 0.4\text{ m} \times 0.3\text{ m}$ in the channel, there is a uniform magnetic field generated by the superconducting coil, with a magnetic induction intensity $\text{B} = 6.4\text{ T}$

and a direction perpendicular to the face of area $a \times c$ outward. On the magnetic field area, there are metal plates M and N with area $(a \times b = 0.3\text{ m} \times 0.4\text{ m})$. When they are connected to the dedicated DC power supply for the thruster, a wave from N to M is generated in the seawater between the two plates, and the current is 1000 A through each channel, assuming that this current only exists in the magnetic field area. Excluding the internal resistance of the power supply and the resistance of the wire, and taking the density of seawater $\rho_m = 1000\text{ kg}/\text{m}^3$:-



34. Mark the correct statement.

(1) Changing the magnitude of current in left propeller only can turn the submarine.

(2) Changing the direction of current in left propeller only (keeping magnitude constant) can turn the submarine.

(3) Changing the direction of current in both the propellers can reverse the submarine.

(4) Changing the magnitude of magnetic field in both the propellers can change the speed of submarine.

A) 1 only B) 1 and 2 C) 2 and 3 D) 1, 2, 3 and 4

35. When the submarine moves at a constant speed of $30\text{ m}/\text{s}$, the power supplied by the DC source is :-

A) 67×10^5 W B) 5.5×10^5 W C) 11.17×10^5 W D) 56×10^5 W

SECTION - 4 (Maximum Marks : 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**

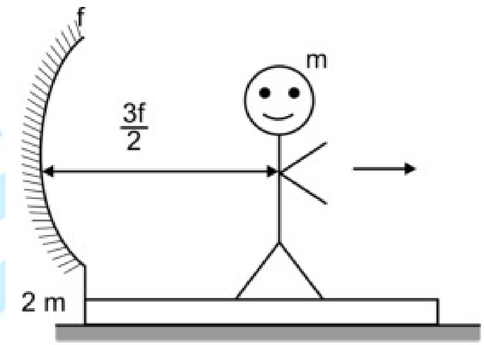
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 If ONLY the correct numerical value is entered as answer.

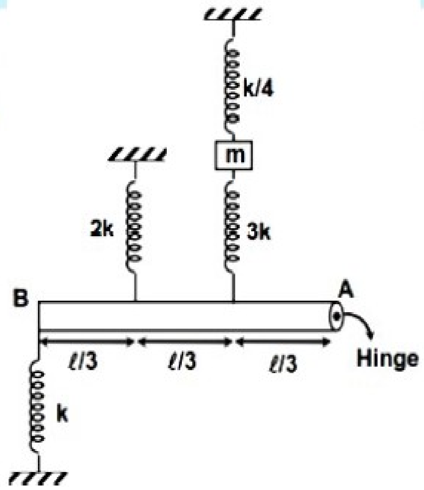
Zero Marks: 0 In all other cases.

36. A man of mass m starts moving w.r.t a platform of mass 2 m with a velocity $u = 9/13$ m/s towards right as shown in the figure. The platform is fitted with a concave mirror of focal length f. The velocity of image (in m/s) at the moment is :-



37. A person is observing a stone at the bottom of the pool. The water depth is $h = 14$ m. The angle between the line of sight and the normal line of the water surface is 60°, the refractive index of water is 1.33. The depth of the image of the stone (in m) is :-

38. A thin horizontal light rod AB of length $'l'$ hinged at end A is connected with the ideal vertical springs and a block of mass 'm' in gravity free space as shown in the figure. The block is slightly displaced from its equilibrium position in the plane contain springs and the rod and then released. If the frequency of small oscillations of the block is $\frac{1}{2\pi}\sqrt{n\left(\frac{7k}{20m}\right)}$. Find the value of n.



CHEMISTRY MAX.MARKS: 60

SECTION - 1 (Maximum Marks : 24)

This section contains **SIX (06)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

39. Select the correct options

A) Emulsion can be diluted with any amount of the dispersion medium

B) Emulsions can be broken into constituent liquid by heating, freezing, centrifuging etc.

C) Emulsifying agents for W/O emulsion are proteins, gums, natural and synthetic soaps etc

D) Butter and Cream are W/O type emulsions

40. Select the correct options

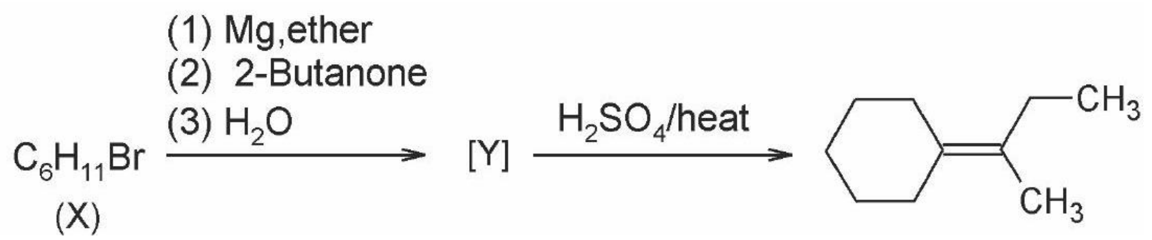
A) Vapour pressure of a pure liquid is an intentive property and also a colligative property.

B) Solution showing very little positive deviation can form minimum boiling point azeotrope

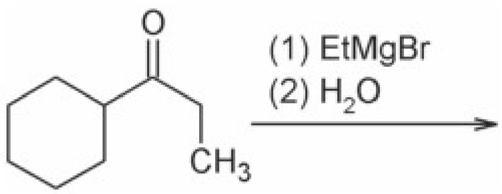
C) Life time for zero order reaction is finite and zero order reaction are always complex in nature.

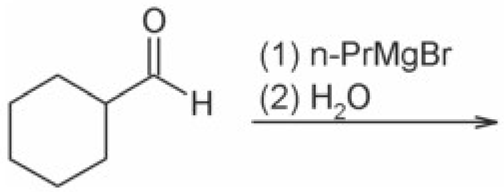
D) Electrolysis product of brine (aq. conc. $NaCl$) using inert electrodes are H_2 and Cl_2 gas at cathode and anode respectively.

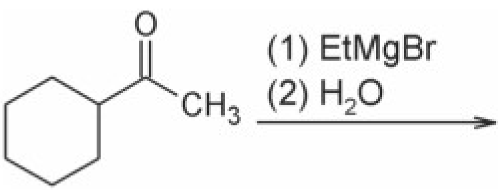
41. Reaction scheme:

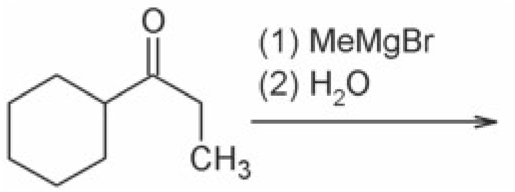


Among the following reactions, Identify the reaction(s) which gives Compound (Y) as a major product?

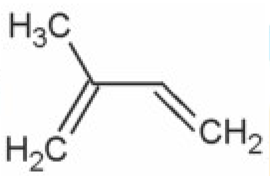
A) 

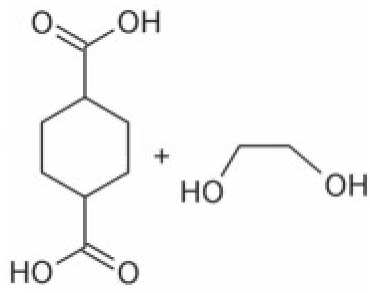
B) 

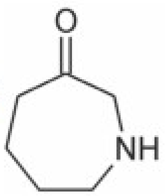
C) 

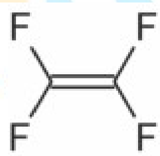
D) 

42. Some name of polymers and corresponding monomers are given below. Identify the correctly matched option(s)?

A) Natural rubber: 

B) Dacron: 

C) Nylon-6: 

D) Teflon: 

43. How many among the following groups are of lewis acids

A) $BF_3, BCl_3,\ B_2H_6$

B) H^+, NH_4^+, CH_4

C) $BCl_3, AlCl_3,\ B_2H_6$

D) $CH_3CO^+, Fe^{3+}, Mg^{2+}$

44. Correct statements about silicones is/are-

A) These are organosilicon polymers

B) A famous example of silicones is zeolite

C) The chain length of the polymer can be controlled by adding $(CH_3)_3SiCl$

D) Cu is used as catalyst in preparation of $(CH_3)_2SiCl_2$ from $CH_3Cl + Si$

SECTION - 2 (Maximum Marks : 12)

This section contains **TWO** (**3**) Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +2 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 45 and 46:

An inorganic compound (A) reacts with dilute H_2SO_4 to give a colourless gas (B) which smells like rotten egg.

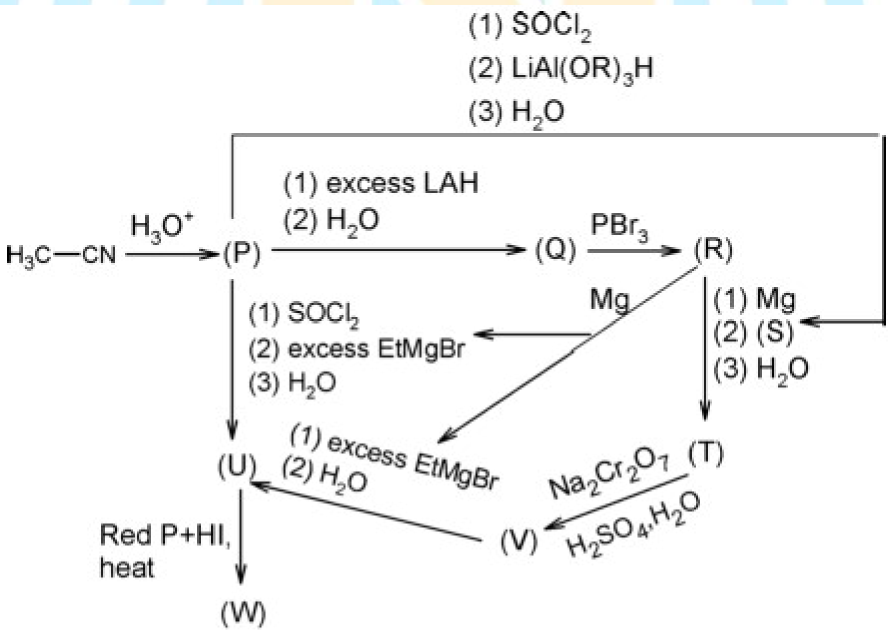
Black precipitate of $(X) \xleftarrow{AgNO_3} B \xrightarrow{Alk.Na_2[Fe(CN)_5NO]}$ Violet coloured compound (Y)

45. Change in oxidation state of Fe, When $Na_2[Fe(CN)_5NO]$ reacts with B to give (Y).

46. Number of moles of acidified $K_2Cr_2O_7$ required to oxidise 340 gram gas B.

(Given atomic mass :
$H=1, O=16, C=12, F=19, N=14, S=32, P=31, Cl=35.5, Br=80, I=127$,
$K=39, Na=23, Pb=207, Al=27, Si=28, Mn=55, Cr=52$)

Question Stem for Question Nos. 47 and 48:



47. Total number of compounds from (P) To (U) which can give yellow ppt when react with NaOI ?

48. Total number of Monochlorinated products obtained when compound(W) undergoes photochlorination?

Question Stem for Question Nos. 49 and 50:

The unit cell is the smallest repeating unit in a crystal structure. The unit cell of gold crystal is found by X-ray diffraction to have the face-centred cubic unit structure. The side of the unit cell is found to be $0.400\ \text{nm}$. The density of Au is $2.0625 \times 10^4\ \text{kg m}^{-3}$

49. If mass of one Au atom is $X \times 10^{-25}\ \text{kg}$. Then value of X will be

50. If the Avogadro constant is $Y \times 10^{23}$, then value of Y is

(Given that the relative atomic mass of Au is 198.33)

SECTION - 3 (Maximum Marks : 12)

This section contains **TWO** (**2**) Paragraphs. Based on each paragraph, there are **2** questions.

Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer. For each question, choose the option corresponding to the correct answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.
Negative Marks: −1 In all other cases.

Question Stem for Question Nos. 51 and 52:

When $\left[Co(NH_3)_4(H_2O)_2\right]^{3+}$ reacts with excess of NH_3, $\left[Co(NH_3)_5(H_2O)\right]^{3+}$ is formed then HCl in added to form a complex (X) which contain no H_2O as a ligand & shows purple colour

51. If one mole of (X) gives two moles of $AgCl$ on treating with excess $AgNO_3$ solution then its formula and its magnetic nature is :-

A) $CoCl_3.6NH_3$, diamagnetic B) $CoCl_3.5NH_3$, diamagnetic

C) $CoCl_3.4NH_3$, paramagnetic D) $CoCl_3.5NH_3$, paramagnetic

52. Select correct about this complex (X) :-

A) It is an inner orbital octahedral complex.

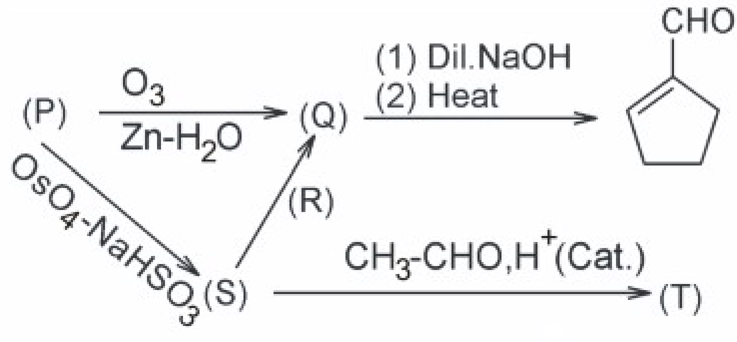
B) Complex show geometrical isomerism

C) IUPAC name of complex is tetraaminedichloridocobalt(III) chloride

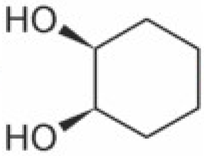
D) Van't Hoff factor of complex $= 2$

Question Stem for Question Nos. 53 and 54:

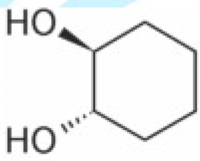
Reaction scheme:



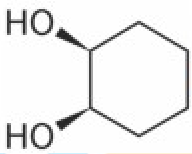
53. Identify Compound(S) and Reagent(R) in above reaction scheme?

A) Compound(S) = 

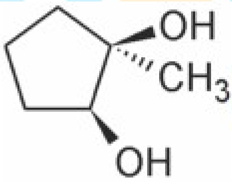
Reagent(R) = P.C.C.(Pyridinium chlorochromate)

B) Compound(S) = 

Reagent (R)= Pb $(OAc)_4$

C) Compound(S) = 

Reagent (R)= HIO_4

D) Compound(S) = 

Reagent(R) = HIO_4

54. Correct statement about compound (T) is?

A) It is Chiral compound.

B) It has 3 Chiral center

C) It can be classified as Ketal

D) It can reduce fehling solution

SECTION - 4 (Maximum Marks : 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**

For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

55. ΔG (in Joule) for given transformation is '9x'. then the value of 'x' is_____

5 moles $H_2O(\ell, 1\text{ bar}, T\text{ K}) \rightarrow H_2O(\ell, 10\text{ bar}, T\text{ K})$

(density of water $= 1\text{gm/mL}$ & $1\text{ lit - bar} = 100\text{ J}$)

56. EMF of the given galvanic cell at $25^\circ C$

$Pt\,|\,H_2(1\text{ bar}$ $Pt\left|H_2(1\text{bar})\right|RNH_3^+Cl^-(1M)\left\|CH_3COOH(0.1M)\right|H_2(1\text{bar})\left|Pt\right.$

is 0.12 V, then pK_b for RNH_2 (weak base) will be

Given that: $K_a(CH_3COOH) = 10^{-5}$, and $\frac{2.303RT}{F} = 0.06$.

57. For the given compound, the total number of optically active stereoisomers is?